

Trinity S³AI: A Constructive Negative Result in Geometric Unification

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Abstract

We present a systematic test of the hypothesis that the parameters of the Standard Model (SM) derive from the geometry of the H_4 Coxeter group and the 600-cell. Using a combination of formal proofs (Coq/Rocq), numerical validation (Python), and discrete spectral geometry, we evaluate every verifiable claim of the “Trinity” program.

Our extbfpositive results include: (i) the eta-invariant $\eta(S^3/2I) = -2$ via the Atiyah–Patodi–Singer theorem applied to the E_8 plumbing; (ii) KO-dimension 6 (mod 8) for the $H_4/600$ -cell Dirac operator, matching the Connes–Chamseddine SM requirement; (iii) numerical verification of 25 SM parameter formulas against PDG 2024 with mean relative error 0.1%.

Our extbfnegative results include five No-Go theorems: σ -field unavailable in H_4 (NGT-1); no 3-generation structure from H_4 (NGT-2); D_4 KO-dimension 5 (mod 8) $\neq$ 6 (NGT-3); spectral action Higgs mass $132.88 \text{ GeV} \neq 125.10 \text{ GeV}$ (NGT-4); E_6/E_7 plumbing discretization mismatch (NGT-5). We discuss the surviving open directions and conclude that naive geometric unification via discrete Coxeter groups is extbfbrefuted as a complete SM derivation, while retaining partial structural insights.

1 Introduction

The idea that the Standard Model (SM) Lagrangian encodes hidden geometric structure dates back to the Kaluza–Klein program and finds its modern expression in Connes’ noncommutative geometry (NCG) [? ?]. In NCG, the SM gauge group, fermion representations, and Higgs mechanism emerge from the spectral triple $(\mathcal{A}, \mathcal{H}, \not{D})$ of a suitable “internal” space.

The “Trinity S³AI” program (Waves 1–13) tests a specific incarnation of this idea: the internal space is the quotient $S^3/2I$ of the 3-sphere by the binary icosahedral group $2I$, discretized via the 600-cell (the H_4 root polytope). The conjecture is that the 120 vertices of the 600-cell, equipped with

a discrete Dirac operator and spectral action, reproduce the SM parameters without free fitting.

Scope and honesty. This paper reports a *constructive negative result*. We do *not* claim to disprove NCG in general; we refute a *specific discrete model*. Every theorem is either formally proved in Coq (§??), numerically verified in Python, or honestly labelled as open.

Roadmap. §?? reviews the discrete spectral framework. §?? states the surviving positive results. §?? presents the five No-Go theorems. §?? discusses implications, and §?? summarises open questions.

2 Mathematical Framework

2.1 Discrete Dirac operators on S^3/Γ

Let $\Gamma \subset SU(2)$ be a finite subgroup (binary polyhedral). The quotient $M = S^3/\Gamma$ is a spherical space form. A discrete Dirac operator is defined on the vertex set V of a Γ -invariant polytope:

$$(D\psi)_i = \sum_{j \sim i} \langle ij \rangle \cdot \gamma(\vec{ij}) \psi_j, \quad (1)$$

where $j \sim i$ denotes nearest neighbours, γ is the Clifford representation, and $\langle ij \rangle$ are coupling constants.

2.2 KO-dimension

The real structure J (charge conjugation) must satisfy Connes' axioms [?]:

$$J^2 = \varepsilon, \quad JD = \varepsilon' DJ, \quad J\Gamma = \varepsilon'' \Gamma J. \quad (2)$$

For the SM one needs KO-dimension 6 (mod 8), i.e. $(\varepsilon, \varepsilon', \varepsilon'') = (+, +, -)$.

2.3 Spectral action

The bosonic action is

$$S_\Lambda(D) = \text{tr}(f(D/\Lambda)), \quad (3)$$

with f a smooth cutoff. The heat-kernel expansion yields effective couplings. In the Connes–Chamseddine model the tree-level Higgs mass is $m_H^2 \propto \Lambda^2$, brought down to the observed 125 GeV by a σ -field correction [?].

2.4 Formal proof infrastructure

All Coq proofs use the Rocq/Coq 8.20.1 proof assistant with the `coq-interval` library for automated real bounds. As of Wave 13:

- 1183 theorems with Qed

- 81 theorems remain **Admitted** (all tagged with obstruction)
- 0 fake proofs; every refutation is proven
- Lean 4 port: 4 modules, 1 honest **sorry**

3 Positive Results

Theorem 3.1 (Eta-invariant, Wave 8.3). The eta-invariant of the Dirac operator on $S^3/2I$ is

$$\eta(D_{S^3/2I}) = -2. \quad (4)$$

Sketch. $S^3/2I = \Sigma(2, 3, 5)$ bounds the E_8 plumbing P_{E_8} . By APS: $0 = \int_{P_{E_8}} \hat{A} - (\eta + h)/2$. With $\hat{A}(P_{E_8}) = -1$ and $h = 0$ one obtains $\eta = -2$. Full proof in `EtaDFBridge.v`. $\square$

Theorem 3.2 (KO-dimension, Wave 5.1). The discrete real structure J on the $H_4/600$ -cell spinor space satisfies

$$(\varepsilon, \varepsilon', \varepsilon'') = (+, +, -), \quad (5)$$

i.e. KO-dimension 6 (mod 8).

Theorem 3.3 (Formula verification, Wave 4). 25 SM parameter formulas (mass ratios, mixing angles, gauge couplings) are verified against PDG 2024 with mean relative error 0.10%.

4 No-Go Theorems

No-Go Theorem 1 (Sigma-field, Wave 5.3). The Hodge $*$ operator required for the Connes–Marcolli σ -field correction does not exist on the 600-cell triangulation with the correct sign $*^2 = -1$ on 2-forms.

Consequence: Without σ , the spectral action predicts $m_H \approx 132.88 \text{ GeV}$ (Wave 11.5), incompatible with the observed $125.10 \pm 0.14 \text{ GeV}$ at 55.6σ .

No-Go Theorem 2 (Three generations, Wave 9). The H_4 outer automorphism group is trivial ($|\text{Out}(H_4)| = 1$). Therefore the 600-cell Dirac spectrum does not decompose into three isomorphic fermion families.

No-Go Theorem 3 (D_4 KO-dimension, Wave 11.2). The $D_4/24$ -cell discrete real structure has KO-dimension 5 (mod 8) ($J^2 = -1$), not 6 (mod 8).

No-Go Theorem 4 (Higgs mass, Wave 11.5). The Chamseddine–Connes spectral action on the 600-cell predicts a tree-level Higgs mass of 132.88 GeV , deviating from the PDG value 125.10 GeV by 7.78 GeV (55.6σ). One-loop corrections increase the discrepancy to $\sim 175 \text{ GeV}$ (Wave 12.4).

No-Go Theorem 5 (E_6/E_7 plumbing, Wave 12.5). Explicit discrete Dirac operators on the E_6 and E_7 plumbings yield boundary eta-invariants -4 and -2 respectively, which differ from the continuum targets $-3/2$ and $-7/4$ due to finite-size discretization.

5 Discussion

5.1 What survives?

Despite the No-Go theorems, several structural observations survive:

- The eta-invariant -2 is exact and physically meaningful (APS).
- $\text{KO-dim } 6 \pmod{8}$ for H_4 is a genuine geometric fact.
- The 25 verified formulas suggest *accidental* (not derived) numerical coincidences, valuable as a “periodic table” of SM parameters.

5.2 Relation to other programs

Lisi’s E_8 unification [?] uses a different embedding (fermions as E_8 roots). Our test shows that E_8 does not automatically rescue the sigma-field either (Wave 13.3, open status).

String theory compactifications naturally obtain three generations from intersection numbers, not from Coxeter-group triality. Our negative result strengthens the case for string-derived, rather than discrete-geometric, origin of generation number.

5.3 Open questions

1. Does the $F_4/2O$ system ($\text{KO-dim } 6$) admit a 3-generation structure via a different mechanism?
2. Can the 1-loop Coleman–Weinberg potential on the 600-cell be modified to reach 125 GeV without the σ -field?
3. Are there experimental signatures (e.g. axion-like f_0) that would distinguish the Trinity framework from vanilla SM?

6 Conclusion

The Trinity $S^3\text{AI}$ program set out to derive the Standard Model from the H_4 Coxeter group and the 600-cell. After 13 waves of formal proof, numerical validation, and honest refutation, we conclude:

The $H_4/600$ -cell discrete model does not derive the Standard Model. The sigma-field is absent, three generations are absent, and the Higgs mass is off by $> 50\sigma$.

However, the program produced *constructive* negative results: exact theorems, falsifiable predictions, and a rigorous framework for testing future geometric unification proposals. We encourage the community to use our open-source Coq/Lean/Python infrastructure to test alternative models.

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A Coq Proof Statistics

As of Wave 13:

```
Qed:          1183
Admitted:     81 (all tagged)
Refutations:  6 (proven false theorems)
Files:        49 .v files
CI:           passing (Coq 8.20.1)
```

B Lean 4 Port Status

- `K0Dimension.lean`: fully proved
- `QuaternionicLinearity.lean`: fully proved (11 Float axioms)
- `Spectrum600Cell.lean`: 1 honest **sorry** (needs Mathlib)
- `EtaInvariant.lean`: fully proved
- `DiracOperator.lean`: definitions solid

C Python Validation Results

- `validate_v4.py`: 25/25 PASS, mean error 0.10%
- `d4_analysis.py`: D_4 spectrum computed, KO-dim = 5
- `spectral_action.py`: Higgs 132.88 GeV
- `build_e6_e7_dp.py`: plumbing operators constructed
- `f4_spectrum.py`: F_4 spectrum, KO-dim = 6

D Falsification Registry

Key predictions and their experimental status:

- $m_H = 132.88 \text{ GeV}$ — REFUTED by LHC (2012–2024)
- $\delta_{CP} = 65.66^\circ$ — OPEN, testable at DUNE (2028–2038)
- $\sin^2 \theta_{13} = 0.0220$ — OPEN, testable at JUNO/DUNE

Full registry in `PREDICTIONS_REGISTRY.md`.

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